PROJECT SYNOPSIS

T.Y. Major Project

1. Title of the Project

VisionAid — A Deep Learning-Based Assistive System for Visually Impaired People

2. Introduction

Vision is the main way most people learn about their surroundings. For a visually impaired person, even routine activities such as walking through a corridor, finding a doorway, avoiding an object left on the path, or reading a notice board depend on information that is hard to get without sight. Traditional aids such as the white cane and guide dogs are useful, but they mainly provide information about what is within physical reach. They cannot tell the user what an object is, and they cannot read a sign, a medicine label or a printed notice. In many situations the user therefore has to ask another person, which reduces independence and is not always possible.

Recent progress in computer vision and deep learning offers a practical way to reduce this gap. Deep learning-based **object detection** models can locate and name common objects in an image or video frame, such as a chair, a person, a door or a vehicle. **Optical Character Recognition (OCR)** can detect text in an image and convert it into machine-readable characters, so that signs, labels, notices and documents become readable by software. Since the user cannot depend on a screen, this information must be delivered through a non-visual channel. **Text-to-speech (TTS)** converts the processed information into spoken audio that the user can hear through a speaker or earphones.

A camera is already available in almost every smartphone, and small wearable cameras are inexpensive. A solution built around a camera and software therefore has the potential to be much more affordable than specialised assistive hardware.

The motivation for **VisionAid** is to combine these three capabilities, object detection, OCR and audio feedback, into one simple prototype that helps a visually impaired user become aware of nearby objects and read visible text. VisionAid is intended as a supporting tool that improves environmental awareness and reduces dependence on others for basic information. It is not intended to replace the white cane, guide dogs or human assistance.

3. Objectives

1. To study the challenges faced by visually impaired people in environmental awareness, obstacle identification and reading text in their surroundings.
2. To develop a camera-based assistive system that captures and interprets the user's surroundings using a smartphone or wearable camera.
3. To implement a deep learning-based object detection module for identifying common objects and obstacles.
4. To implement an OCR module for recognising and extracting text from signs, labels, notices and documents.
5. To convert the detected objects and extracted text into clear and understandable audio feedback.
6. To provide information about objects and text in near real time, with attention to processing delay.
7. To design a simple and accessible interface that can be operated with minimal visual interaction.

8. To help reduce the dependence of visually impaired users on other people for basic environmental information.
9. To evaluate object detection and OCR performance under different environmental conditions such as lighting, distance and occlusion.
10. To study the practical usability of the prototype through a small, controlled user evaluation.

4. Problem Statement

Visually impaired individuals face difficulty in understanding their immediate surroundings. They may not be able to identify objects and obstacles in their path, recognise the objects around them, or read printed and displayed text such as room numbers, direction boards, product labels, notices and documents. Obtaining this information usually requires help from another person, which may not always be available and which limits independent movement.

Traditional assistance methods have limitations. A white cane detects obstacles only through physical contact and gives no information about what the obstacle is. Human guidance is not always available, and braille is available only in limited places. Specialised electronic assistive devices can be expensive and may be out of reach for many users. There is therefore a need for an accessible and affordable assistive technology that can interpret visual information and present it in a form that does not depend on vision.

Existing assistive solutions and research often concentrate on individual functions, for example obstacle detection alone or text reading alone. In daily life, however, a user needs both kinds of information: what is around me, and what does the text in front of me say. This project addresses the problem of building an **integrated prototype that combines object detection and OCR with audio feedback** in a single workflow, using a low-cost camera input, so that a visually impaired user can receive useful information about nearby objects and readable text. The system aims to assist and support the user, and it does not claim to guarantee safety or to replace human help.

5. Scope of the Project

Functional Scope

- Camera-based image and video input from a smartphone or wearable camera.
- Deep learning-based object detection for common objects.
- Identification of objects that may act as obstacles, with an approximate indication of their position (left, centre, right) in the frame.
- OCR-based text detection and text extraction from signs, labels, notices and documents.
- Conversion of selected information into spoken audio feedback.
- Basic environmental awareness through combined object and text information.

User Scope

The primary target users are visually impaired individuals who need assistance in understanding their immediate surroundings. The prototype is designed to be used alongside, and not instead of, existing mobility aids.

Technical Scope

- Deep learning and computer vision.
- Object detection using a pretrained and, if required, fine-tuned model.

- OCR for printed text.
- Image and video frame processing.
- Text-to-speech for audio feedback.

Prototype Scope

The initial implementation will focus on a selected set of common indoor and campus-type objects (for example persons, chairs, tables, doors, stairs-related signs where feasible, bags and vehicles), on readable printed text, and on controlled real-world testing environments such as the college building and campus. The exact object list and the languages supported by OCR will be fixed after a review of the available datasets and models.

Limitations / Outside Scope

- The system is an assistive tool and is not a replacement for human guidance, a white cane or a guide dog.
- Object detection performance may vary with lighting, occlusion, distance, motion blur and camera quality.
- OCR performance may vary with font, text size, blur, orientation, background and lighting. Handwritten text is not a primary target.
- The prototype may initially support only a limited set of objects and languages (English first; other languages are optional future work).
- Distance to an object will only be estimated approximately from the size and position of its bounding box; no depth sensor is planned.
- Complete autonomous navigation, route planning or GPS-based guidance is outside the current scope.
- The system will not be described as guaranteeing collision avoidance or personal safety.

6. Literature Review / Existing System

The review below groups the related work into five areas. The papers cited are well-known published works on object detection, OCR and related tools. They are listed in Section 18 and should be re-checked against their original sources by the team before final submission. Additional domain-specific papers on assistive navigation will be added during the literature survey phase.

A. Assistive Technologies for Visually Impaired People

The most common aids are the white cane and guide dogs. Electronic travel aids have also been developed, many of them using ultrasonic or infrared sensors that warn the user when something is close. Such devices are effective for detecting the presence of an obstacle but usually cannot identify what the obstacle is. Camera-based systems add the ability to recognise objects and scenes. Several smartphone applications, such as Microsoft Seeing AI and Be My Eyes, and wearable products such as OrCam MyEye, show that camera-based assistance is practical; some depend on cloud services, human volunteers or dedicated and costly hardware.

B. Deep Learning-Based Object Detection

Modern object detectors are based on convolutional neural networks. Two-stage detectors such as Faster R-CNN [3] give good accuracy, while single-stage detectors such as YOLO [1] and SSD [2] predict object locations and classes in a single pass, which makes them faster and better suited to real-time use. Lightweight backbones such as MobileNets [4] were designed for mobile and embedded devices. These models are commonly trained on large datasets such as MS COCO [5], which includes many everyday

object categories and is a suitable starting point for detecting common objects relevant to indoor and campus environments.

C. OCR-Based Assistive Systems

OCR converts text in an image into digital text. Tesseract [6] is a widely used open-source OCR engine that works well on clean, well-aligned printed text. Reading text in natural scenes (signs, boards, labels) is harder, and deep learning approaches have been proposed for it, including text detection methods such as EAST [8] and recognition methods such as CRNN [7]. Reading aids for visually impaired users commonly apply OCR to documents, product labels and signs, with accuracy depending strongly on image quality, lighting and framing.

D. Audio-Based Assistive Feedback

Because visually impaired users cannot rely on a display, audio is the most natural output channel. Text-to-speech converts text into synthetic speech and is already used in screen readers and reading aids. For assistive systems, the design of the audio message matters as much as its generation: messages should be short, should not be repeated unnecessarily, and should not overload the user. Audio output should also not block the user's ability to hear the environment, so earphones that allow ambient sound or a small speaker may be considered.

E. Integrated Assistive Systems

Some applications and devices combine several functions, for example scene description, text reading and object finding in one product. Many research prototypes, however, focus on one function, such as obstacle detection, or are developed for a specific environment or hardware platform. Reports on integrated systems that are low-cost, reproducible and evaluated across different environmental conditions are comparatively limited, which motivates the present work.

7. Research Gap

Existing work provides good solutions for object recognition, OCR, assistive navigation and audio feedback individually. However, many solutions concentrate on a single function, depend on specific hardware or controlled environments, or are not easily reproducible at student level with low-cost components.

VisionAid attempts to integrate **Object Detection + OCR + Audio Feedback** into one accessible assistive workflow built with open-source tools and a commonly available camera. It does not claim that this combination is completely new. The contribution of this project is an integrated, student-level prototype, together with a practical evaluation of its behaviour in different conditions and a focus on usability and accessibility, such as prioritising what is spoken to the user.

8. Proposed System / Methodology

VisionAid follows a pipeline that starts with camera input, processes the image through object detection and OCR in parallel, selects the most relevant information and delivers it to the user as speech. The conceptual flow is:

Camera Input -> Image / Video Capture -> Preprocessing -> (Object Detection || OCR) -> Information Processing -> Priority / Relevant Information Selection -> Text-to-Speech -> Audio Feedback

8.1 Camera Input and Capture

A smartphone camera or a small wearable camera (for example, mounted on a cap, chest strap or glasses) captures the user's surroundings. The prototype will work with live video frames and also with still images, so that the system can be tested in a repeatable way.

8.2 Preprocessing

Captured frames are prepared for the detection and OCR modules. Depending on the requirement, this may include resizing to the model input size, normalisation, colour conversion, and mild noise reduction or contrast adjustment for OCR. To keep the system responsive, not every frame needs to be processed; frame skipping or a fixed processing interval will be considered.

8.3 Object Detection

A deep learning object detection model identifies common objects and possible obstacles and returns, for each detection, a class label, a confidence score and a bounding box. A lightweight single-stage detector from the YOLO family is the *proposed* implementation choice because of its speed and good support in Python. The final model (and its size) will be selected after comparing speed and accuracy on the available hardware. A pretrained model will be used first, and fine-tuning on a small custom dataset will be done only if the pretrained model is not adequate for the selected objects.

8.4 OCR

The OCR module detects text regions and extracts the characters from signs, labels, notices and documents. Tesseract OCR is the *proposed* engine, supported by OpenCV-based preprocessing such as grayscale conversion and thresholding. Another OCR framework may be considered if Tesseract does not perform adequately on natural scene text. Since OCR is more computationally demanding and not always needed, the prototype may run it only when a text region is likely to be present or when the user requests reading.

8.5 Information Processing and Priority Selection

The raw outputs of both modules cannot be spoken directly, as this would produce too many messages. The information processing stage therefore:

- removes low-confidence detections using a threshold;
- estimates the approximate position of each object (left, centre or right) from its bounding box, and a rough nearness category from the box size;
- gives higher priority to objects likely to be obstacles in the user's path than to objects at the side;
- avoids repeating the same message for consecutive frames;
- cleans the OCR output (removes noise and very short or meaningless strings); and
- builds short natural-language messages, for example "chair ahead" or "sign reads: Exit".

Two operating modes are proposed. In **Awareness mode** the system continuously reports important nearby objects. In **Reading mode** the user triggers text reading, and the system speaks the text detected in the current view.

8.6 Text-to-Speech and Audio Feedback

The selected messages are converted to speech using a text-to-speech library or service and played through the speaker or earphones. Messages are kept short, and urgent object alerts take precedence over longer text reading. The overall aim is to give useful environmental information with minimum visual interaction.

9. Overall Workflow

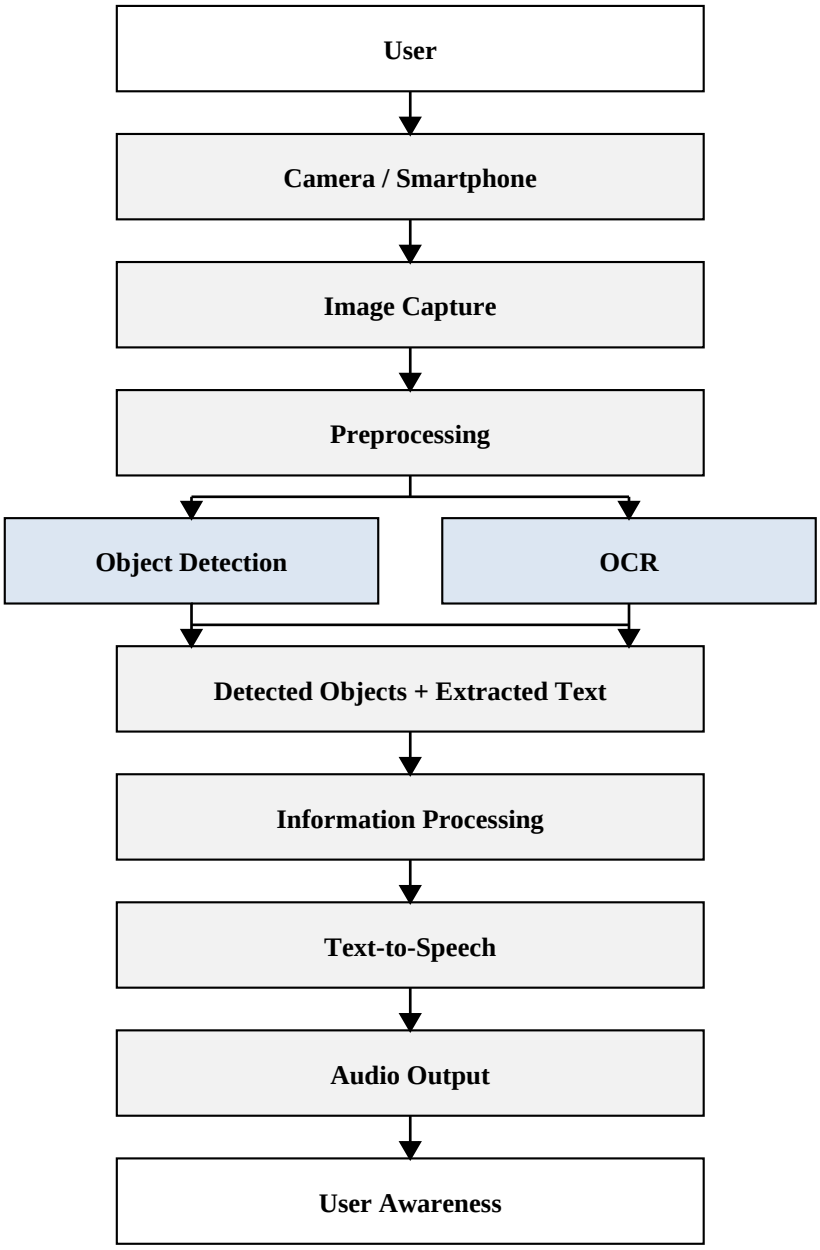

Figure 1: Overall workflow of VisionAid

Step	Workflow Stage	Description
1	Camera Input	Capture surroundings using smartphone / wearable camera
2	Image Processing	Prepare captured frames / images (resize, normalise, enhance)
3	Object Detection	Identify relevant objects and obstacles
4	OCR	Detect and extract visible text
5	Information Processing	Filter, prioritise and organise the relevant detected information
6	Speech Generation	Convert the selected information into audio
7	Audio Feedback	Communicate the information to the user

Step	Workflow Stage	Description
8	User Response	User uses the information for awareness and movement

10. System Architecture

The system is organised into the following layers. Object detection and OCR operate in parallel on the same preprocessed frame.

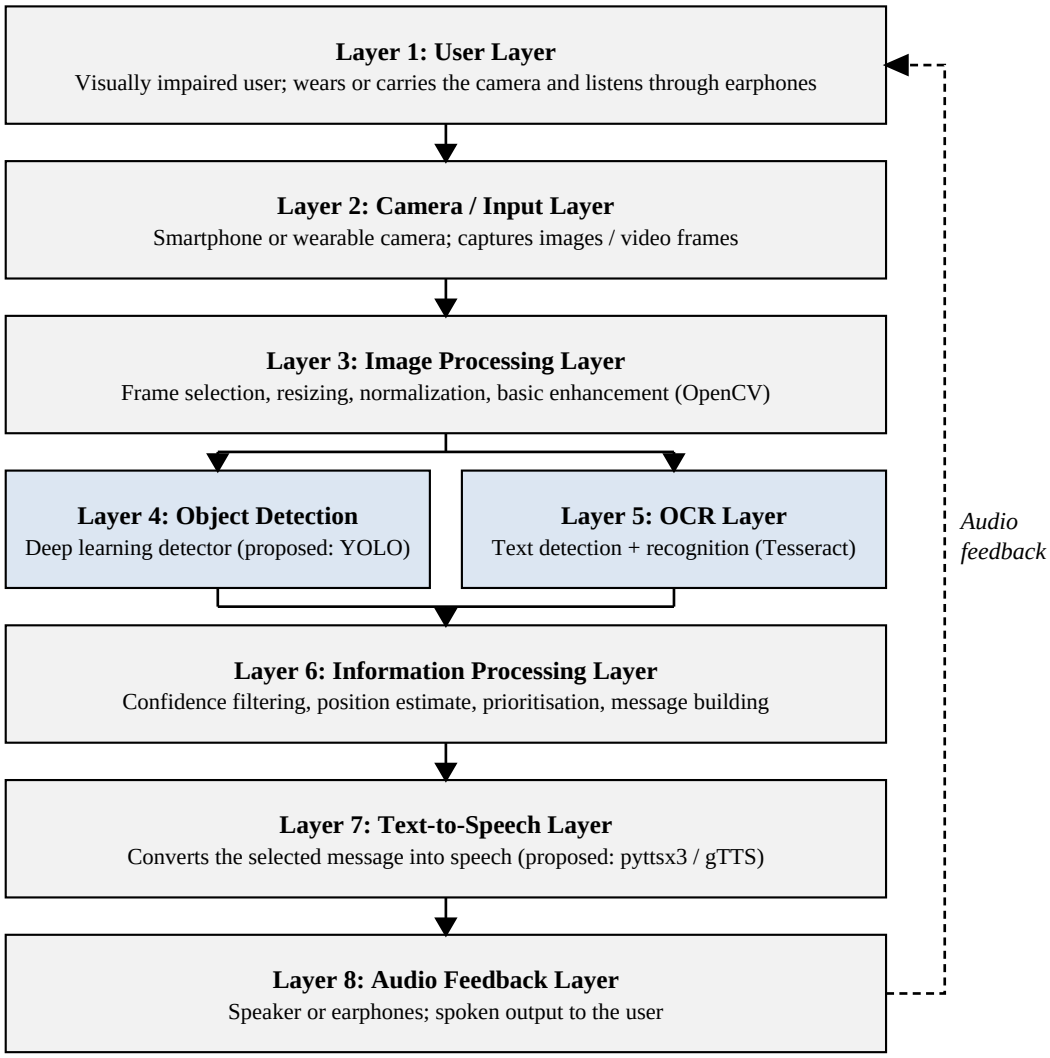

Figure 2: Layered architecture of VisionAid

Layer	Responsibility
1. User Layer	The visually impaired user, who wears or carries the camera, selects the mode and receives audio information.
2. Camera / Input Layer	Captures live video or images of the surroundings and passes frames to the system.
3. Image Processing Layer	Prepares frames: resizing, normalisation, noise reduction and contrast adjustment.

Layer	Responsibility
4. Object Detection Layer	Detects objects and obstacles; outputs labels, confidence scores and bounding boxes.
5. OCR Layer	Finds text regions and recognises text; outputs extracted strings.
6. Information Processing Layer	Filters, prioritises and combines detection and OCR output into short messages.
7. Text-to-Speech Layer	Converts the selected messages into speech.
8. Audio Feedback Layer	Plays the speech through speaker or earphones so that the user can act on it.

11. Proposed Technology Stack

The following technologies are *proposed*. The final choice will be made after initial experiments.

Area	Proposed Technology	Purpose
Programming	Python	Main language for all modules
Computer Vision	OpenCV	Frame capture, preprocessing, drawing detections for testing
Deep Learning	PyTorch (or TensorFlow); YOLO-family lightweight detector	Object detection; fine-tuning if needed
OCR	Tesseract OCR (via pytesseract); alternative OCR framework if required	Text extraction
Audio	Offline TTS library (e.g., pyttsx3) or online TTS (e.g., gTTS)	Speech generation
Interface	Simple Python / web-based prototype interface, with voice or button control	Mode selection and system control
Development	VS Code, Git / GitHub	Coding and version control

12. Implementation Modules

Module 1: Image / Video Capture

Captures input from a smartphone camera, wearable camera or webcam, and provides frames to the rest of the system. It also supports loading stored images and videos for repeatable testing.

Module 2: Object Detection

Detects and identifies the selected objects in each processed frame and outputs class, confidence and location. Includes model selection, optional fine-tuning and threshold tuning.

Module 3: OCR

Preprocesses images for text recognition, extracts visible text and cleans the output. Handles basic variations in size and orientation.

Module 4: Information Processing

Filters, prioritises and organises detected objects and extracted text, removes repeated messages and creates the final short messages to be spoken.

Module 5: Text-to-Speech

Converts the final messages into speech and manages the audio queue so that urgent messages are spoken first.

Module 6: User Interface

Provides a simple interface for starting and stopping the system and switching between Awareness and Reading modes. The interface will use large controls and audio confirmations so that it can be operated with little or no visual interaction.

Module 7: Testing and Evaluation

Evaluates the detection, OCR and complete system under different environmental conditions and carries out the small usability study.

13. Evaluation Plan

This is a planned evaluation. No results are claimed at the synopsis stage; actual results will be reported after the experiments are done.

Object Detection Evaluation

- Precision and recall on a test set of selected objects.
- mAP (mean Average Precision) where ground-truth annotations are available.
- Detection latency and frames per second (FPS) on the target hardware.

OCR Evaluation

- Character accuracy (based on character error rate).
- Word accuracy (based on word error rate).
- Recognition success rate under different conditions (size, blur, orientation, lighting).

System Performance

- End-to-end response time from capture to start of audio.
- Processing latency of each stage.
- Audio generation time.

Environmental Testing

Tests will be repeated under: good lighting; low lighting; different distances; different object sizes; partial occlusion; and different text sizes and orientations.

Usability Evaluation

If feasible, a small user evaluation will be conducted, covering task completion (for example finding and reading a room sign or naming an obstacle), ease of interaction, and usefulness and clarity of the audio feedback. Early testing may be done with blindfolded sighted volunteers in a controlled and safe setting. Testing with visually impaired users, if possible, will be carried out only with their consent, with supervision and with guidance from the project guide or a suitable organisation.

14. Expected Outcomes / Deliverables

1. A working prototype of the assistive system integrating detection, OCR and audio output.
2. Camera-based environmental input using a smartphone, wearable camera or webcam.
3. An object detection capability for a selected set of common objects and obstacles.

4. OCR-based text recognition and extraction from signs, labels and documents.
5. An audio feedback mechanism using text-to-speech.
6. An information-processing component that prioritises and reduces repeated messages.
7. A simple interface with Awareness and Reading modes.
8. A test dataset of images / videos collected under varied conditions for evaluation.
9. An evaluation of detection, OCR and system performance under different environmental conditions.
10. Findings from a small usability study.
11. Experimental results after completion of the planned experiments.
12. Complete project documentation and final project report.

15. Hardware & Software Requirements

Hardware Requirements

Component	Requirement
Development Computer	Laptop / desktop capable of running Python, OpenCV and a deep learning model (GPU is helpful but not mandatory)
Camera	Smartphone camera, wearable camera or webcam for capturing surroundings
Audio Output	Earphones or a small speaker
Storage	Sufficient storage for datasets, test images / videos and model files
Mounting (optional)	Cap, chest strap or glasses mount to hold the camera during testing

Software Requirements

Software	Purpose
Python and required libraries	Implementation of all modules
OpenCV	Image and video processing
PyTorch or TensorFlow	Running and, if needed, fine-tuning the object detection model
Tesseract OCR	Text recognition
Text-to-speech library	Speech generation
VS Code, Git / GitHub	Development and version control

16. Project Plan / Timeline

Phase	Duration	Activities	Expected Output
Project Planning and Problem Definition	2 Weeks	Problem definition, objectives, scope and deliverables	Approved project direction
Literature Survey	3 Weeks	Study of assistive systems, object detection, OCR and TTS	Literature summary and research gap

Phase	Duration	Activities	Expected Output
Requirement Analysis and System Design	3 Weeks	Requirements, architecture, workflow, selection of tools and models	System design document
Dataset Preparation	2 Weeks	Selection of object classes, collection and annotation of test data	Test dataset
Object Detection Module	4 Weeks	Model setup, testing, optional fine-tuning, threshold tuning	Working detection module
OCR Module	3 Weeks	OCR setup, preprocessing and text cleaning	Working OCR module
Information Processing and TTS	3 Weeks	Prioritisation logic, message generation and speech output	Audio feedback pipeline
Interface and Integration	3 Weeks	Simple interface, integration of all modules	Integrated prototype
Testing and Evaluation	3 Weeks	Performance, environmental testing and usability study	Evaluation results
Documentation	2 Weeks	Final report, presentation and demonstration	Final documentation

17. Conclusion

VisionAid proposes a deep learning-based assistive prototype that helps visually impaired users become aware of their surroundings. By using a smartphone or wearable camera, detecting objects and obstacles, reading visible text with OCR and giving the information as audio, the system aims to support independent movement and reduce dependence on other people for basic environmental information. The project will use open-source tools and low-cost hardware, and its performance will be evaluated under varied conditions. VisionAid is intended as an assistive aid, and it is not a replacement for human guidance or traditional mobility aids. It does not guarantee safety. The planned work is a practical, integrated student-level prototype that contributes towards more accessible and affordable assistive technology.

18. References

The following are published works relevant to the proposed system. They are included as additional references from general knowledge and must be verified by the team against the original sources before submission.

1. J. Redmon, S. Divvala, R. Girshick and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," IEEE CVPR, 2016.
2. W. Liu et al., "SSD: Single Shot MultiBox Detector," ECCV, 2016.
3. S. Ren, K. He, R. Girshick and J. Sun, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks," NeurIPS, 2015.
4. A. G. Howard et al., "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications," arXiv:1704.04861, 2017.
5. T.-Y. Lin et al., "Microsoft COCO: Common Objects in Context," ECCV, 2014.
6. R. Smith, "An Overview of the Tesseract OCR Engine," Proc. International Conference on Document Analysis and Recognition (ICDAR), 2007.

7. B. Shi, X. Bai and C. Yao, "An End-to-End Trainable Neural Network for Image-Based Sequence Recognition and Its Application to Scene Text Recognition," IEEE TPAMI, 2017.
8. X. Zhou et al., "EAST: An Efficient and Accurate Scene Text Detector," IEEE CVPR, 2017.

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